Zhenya Liu

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Research interests: Reinforcement learning, with applications to robotic policy learning and LLM post training. I develop **simple and effective** RL algorithms grounded in theory and useful in general practice.

EDUCATION

University of Chicago — Chicago, IL
Ph.D. in Computer Science; Advisor: Yuxin Chen — Sep 2024 to Jan 2028 (expected)
M.S. in Computational and Applied Mathematics — 2022 to 2024

University of Wisconsin Madison — Madison, WI
Bachelor's degree in Mathematics — 2018 to 2021
GPA: 3.83/4.00 | Dean's List (all undergraduate semesters)

EXPERIENCE

PerfectBit, Inc. | Robotics Research Intern — Sep 2026 to Present
Founded by Peter Vajda (former Meta Media Generation Director) and Seiji Yamamoto (former lead of Core Llama teams at Meta).

- Joined as the company's **first intern**; developing **new methods for fine tuning flow matching policies** to solve specific robotic manipulation tasks.
- Building **vision language action (VLA) systems** and **agentic robotics** programming to translate practical operational requirements into simulated tasks, environments, and rewards to train robotic arms for specified behaviors.

SELECTED RESEARCH & PUBLICATIONS

Active Curriculum Refinement for Reinforcement Learning
Zhenya Liu, Yuxin Chen
ICML 2026 — Paper | Code

- Proposed PATH, a **general curriculum refinement framework** that uses active learning over prerequisite graphs to select informative sequences of training tasks rather than isolated environments.
- Designed and implemented a transition from broad exploration to targeted refinement, directing training toward tasks the agent has not yet mastered to improve **generalization to unseen environments**.

Shared Actors Need Not Share Critics: Effects of Value Mismatch in Parallel Reinforcement Learning
Zhenya Liu, Yang Meng, Zhuokai Zhao, Xuefeng Liu, Yuxin Chen
NeurIPS 2026 DynaFront — Paper | Code

- Challenged the shared critic default: designed and implemented **environment conditioned value estimation** to resolve conflicting value targets across environments while keeping the policy unchanged.
- Established theory in deterministic bandits explaining why identical expected policy gradients can produce different learning dynamics; collaborated with a Meta AI researcher and achieved **40.8% higher aggregate normalized return** than a shared critic on unseen levels across 16 games.

Rethinking Transfer in Continual Learning: A Replay Based Realisation
Yang Meng, **Zhenya Liu**, Zhuokai Zhao, Yuxin Chen
NeurIPS 2026 CL4FMAgents — Paper | Code

- Contributed the core idea of Transfer Selective Replay: **use past experience to improve new task learning**, not just prevent forgetting, through compatibility based selection and distillation for stability.
- Designed experiments on continual adaptation of language models; the method was the only evaluated approach to **improve average performance on earlier tasks** across all tested settings.

TECHNICAL SKILLS

Programming & tools: Python, PyTorch, NumPy, Bash, Git, Linux, SLURM.
Core expertise: Reinforcement learning, LLM post training, embodied AI, theoretical analysis.